

## 단계1: 네트워크 생성

```
docker network create my-net  
docker network ls
```

```
good593 🕶️ ➤ ~/dev/docker/test  
▶ docker network create my-net  
0c34315c062fd9ebaead496a57dfd75902ee5b1d1ce360696f6c1ad5fef7f247  
good593 🕶️ ➤ ~/dev/docker/test  
▶ docker network ls  
NETWORK ID          NAME                DRIVER              SCOPE  
90776f53ad48        bridge             bridge              local  
1d7750c43d8e        host              host               local  
0c34315c062f        my-net            bridge             local  
32eab69fc42d        none              null               local  
good593 🕶️ ➤ ~/dev/docker/test  
▶ █
```

## 단계2: database 디렉토리 생성

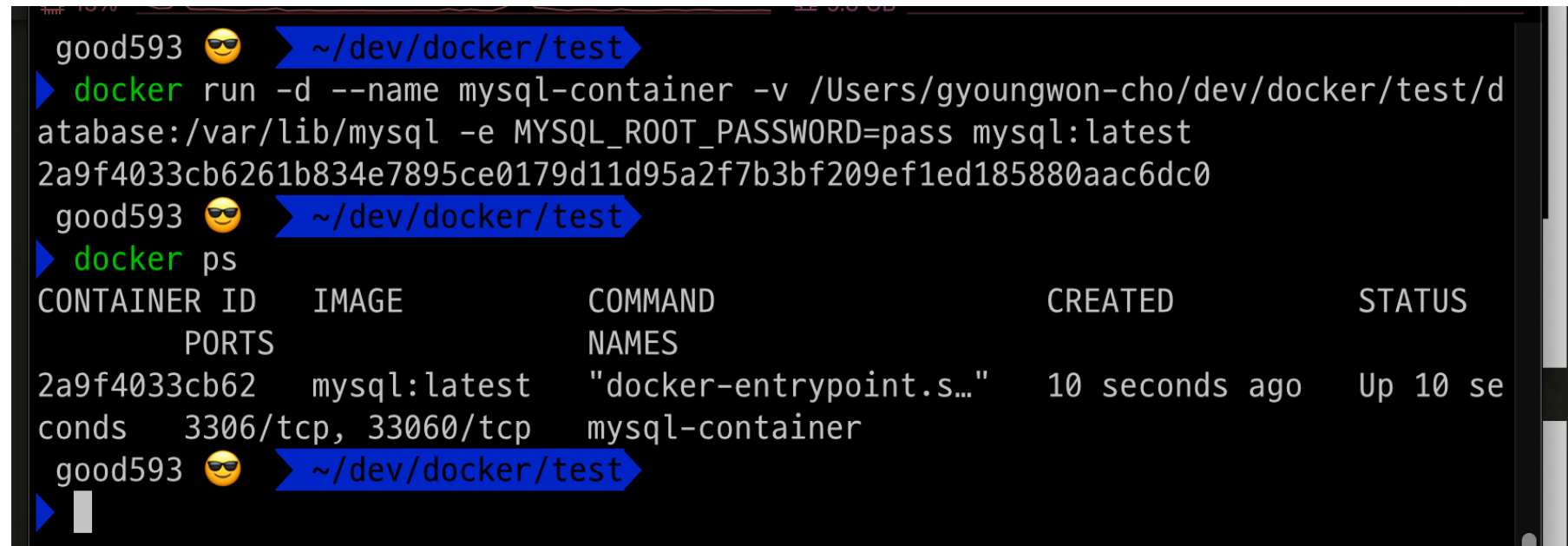
1. 테스트할 폴더로 이동
2. database 디렉토리(폴더) 생성

```
good593 🕶️ ➡ ~/dev/docker/test
> mkdir database
good593 🕶️ ➡ ~/dev/docker/test
> ls
database
good593 🕶️ ➡ ~/dev/docker/test
> pwd
/Users/gyoungwon-cho/dev/docker/test
good593 🕶️ ➡ ~/dev/docker/test
> █
```

## 단계3: mysql 컨테이너 생성 및 실행

```
docker run -d --name mysql-container \
-v /Users/gyoungwon-cho/dev/docker/test/database:/var/lib/mysql \
-e MYSQL_ROOT_PASSWORD=pass mysql:latest
```

```
docker ps
```



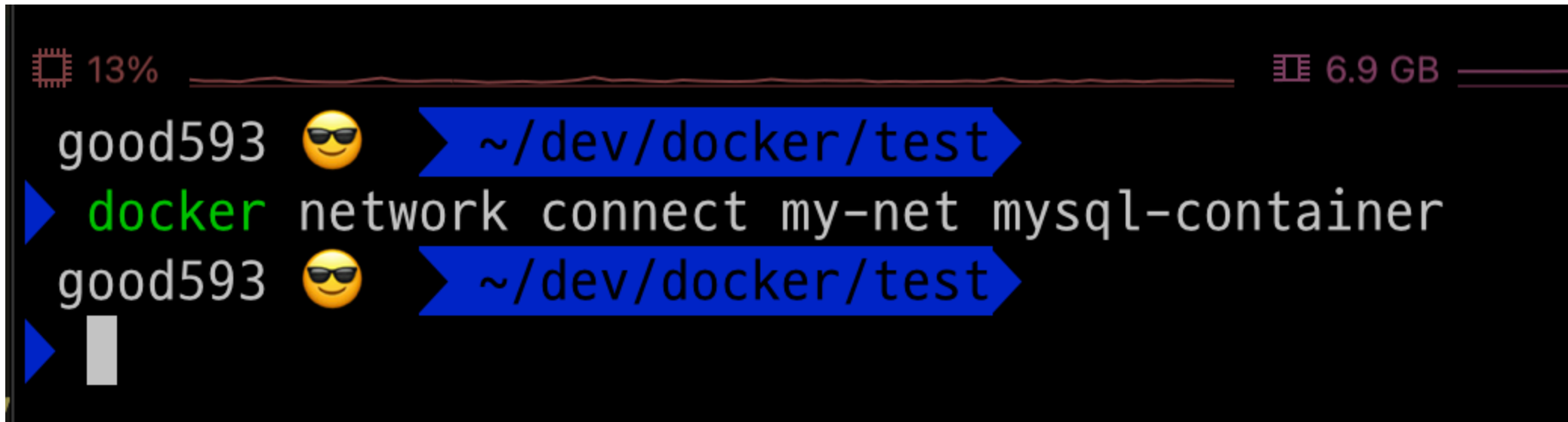
```
good593 🕶 ~/dev/docker/test
> docker run -d --name mysql-container -v /Users/gyoungwon-cho/dev/docker/test/d
atabase:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=pass mysql:latest
2a9f4033cb6261b834e7895ce0179d11d95a2f7b3bf209ef1ed185880aac6dc0
good593 🕶 ~/dev/docker/test
> docker ps
```

| CONTAINER ID | IMAGE        | COMMAND                  | CREATED        | STATUS   |
|--------------|--------------|--------------------------|----------------|----------|
| 2a9f4033cb62 | mysql:latest | "docker-entrypoint.s..." | 10 seconds ago | Up 10 se |

```
cond 3306/tcp, 33060/tcp mysql-container
good593 🕶 ~/dev/docker/test
> █
```

## 단계4: mysql > 네트워크 연결

```
docker network connect my-net mysql-container
```



A terminal window with a dark background. At the top left, there is a CPU icon followed by '13%'. At the top right, there is a memory icon followed by '6.9 GB'. The terminal shows a session with a user named 'good593' who has a sunglasses emoji next to their name. The prompt is '~ /dev/docker/test'. The user enters the command 'docker network connect my-net mysql-container'. The command is highlighted with a blue arrow pointing to the right. Below the command, there is a grey rectangular bar.

```
good593 🕶 ~ /dev/docker/test  
▶ docker network connect my-net mysql-container  
good593 🕶 ~ /dev/docker/test  
▶
```

## 단계5: pymysql 이미지 생성

- Dockerfile이 있는 폴더로 이동 후 실행

```
docker build -t pymysql:latest .
```

```
good593 ♥ ~/dev/github/course_dev_basic/docker/samples/5. 컨테이너간 통신 예제 ▶ dev ±
▶ docker build -t pymysql:latest .
[+] Building 3.1s (11/11) FINISHED
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 321B 0.0s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:18.04 2.9s
=> [auth] library/ubuntu:pull token for registry-1.docker.io 0.0s
=> [1/5] FROM docker.io/library/ubuntu:18.04@sha256:152dc042452c496007f07ca9127571cb9c29697 0.0s
=> [internal] load build context 0.0s
=> => transferring context: 224B 0.0s
=> CACHED [2/5] RUN apt-get -y update && apt-get -y upgrade && apt-get -y dist-upgr 0.0s
=> CACHED [3/5] RUN pip3 install --upgrade pip && pip3 install pandas PyMySQL 0.0s
=> [4/5] COPY main.py /root/ 0.0s
=> [5/5] WORKDIR /root 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:ed00b97fb30ff16676ce456aa9a00176f83175700f7f7cf0908aa9c0b320dd88 0.0s
=> => naming to docker.io/library/pymysql:latest 0.0s
good593 ♥ ~/dev/github/course_dev_basic/docker/samples/5. 컨테이너간 통신 예제 ▶ dev ±
```

## 단계6: pymysql 컨테이너 생성/실행 및 네트워크 연결

```
docker run -d --name pymysql-container --network=my-net pymysql
```

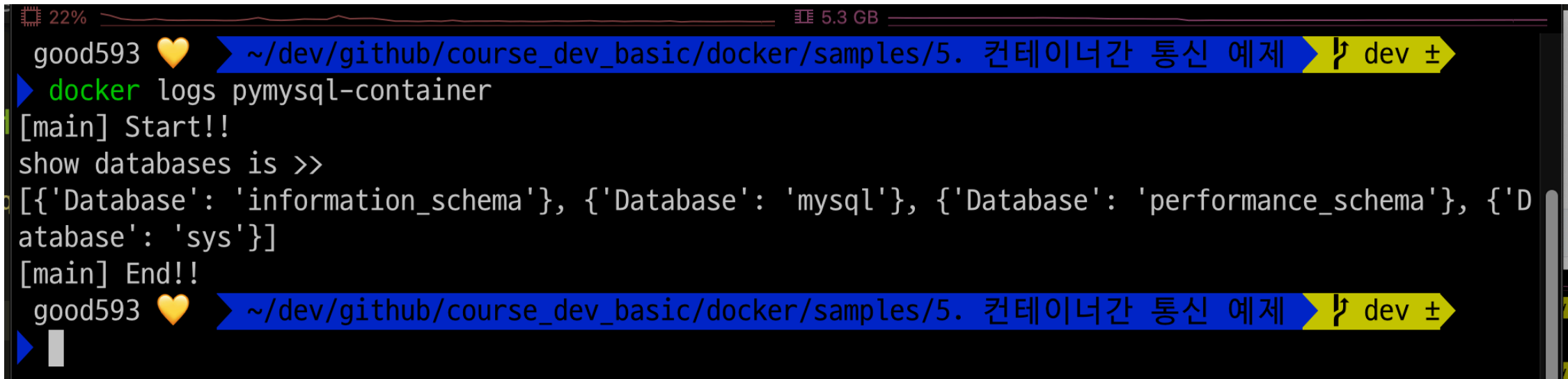


A terminal window with a dark background. The top status bar shows '13%' on the left and '6.4 GB' on the right. The terminal content shows a user 'good593' with a heart icon typing a command. The command is highlighted in blue. The output is a long alphanumeric string. Below the output, the user types another command, also highlighted in blue. The terminal has a blue prompt character on the left.

```
good593 ♥ ~/dev/github/course_dev_basic/docker/samples/5. 컨테이너간 통신 예제 ↗ dev ±  
▶ docker run -d --name pymysql-container --network=my-net pymysql  
045090406b757b7d98022e2a58a17808daa307ad07eff4cfb1a21890b295dd09  
good593 ♥ ~/dev/github/course_dev_basic/docker/samples/5. 컨테이너간 통신 예제 ↗ dev ±  
▶
```

## 단계7: main.py > 결과 확인

```
docker logs pymysql-container
```

A terminal window with a dark background. The top status bar shows '22%' on the left and '5.3 GB' on the right. The prompt is 'good593' followed by a yellow heart icon and a blue path '~ /dev/github/course\_dev\_basic/docker/samples/5. 컨테이너간 통신 예제' with a yellow arrow pointing to 'dev ±'. The command 'docker logs pymysql-container' is entered. The output shows '[main] Start!!', 'show databases is >>', a list of databases in JSON format, '[main] End!!', and the prompt again.

```
good593 ♥ ~ /dev/github/course_dev_basic/docker/samples/5. 컨테이너간 통신 예제 dev ±
> docker logs pymysql-container
[main] Start!!
show databases is >>
[{'Database': 'information_schema'}, {'Database': 'mysql'}, {'Database': 'performance_schema'}, {'Database': 'sys'}]
[main] End!!
good593 ♥ ~ /dev/github/course_dev_basic/docker/samples/5. 컨테이너간 통신 예제 dev ±
> 
```